

AI-Driven Non-Invasive Blood Glucose Forecasting Using Multimodal Physiological Signals and Dietary Data



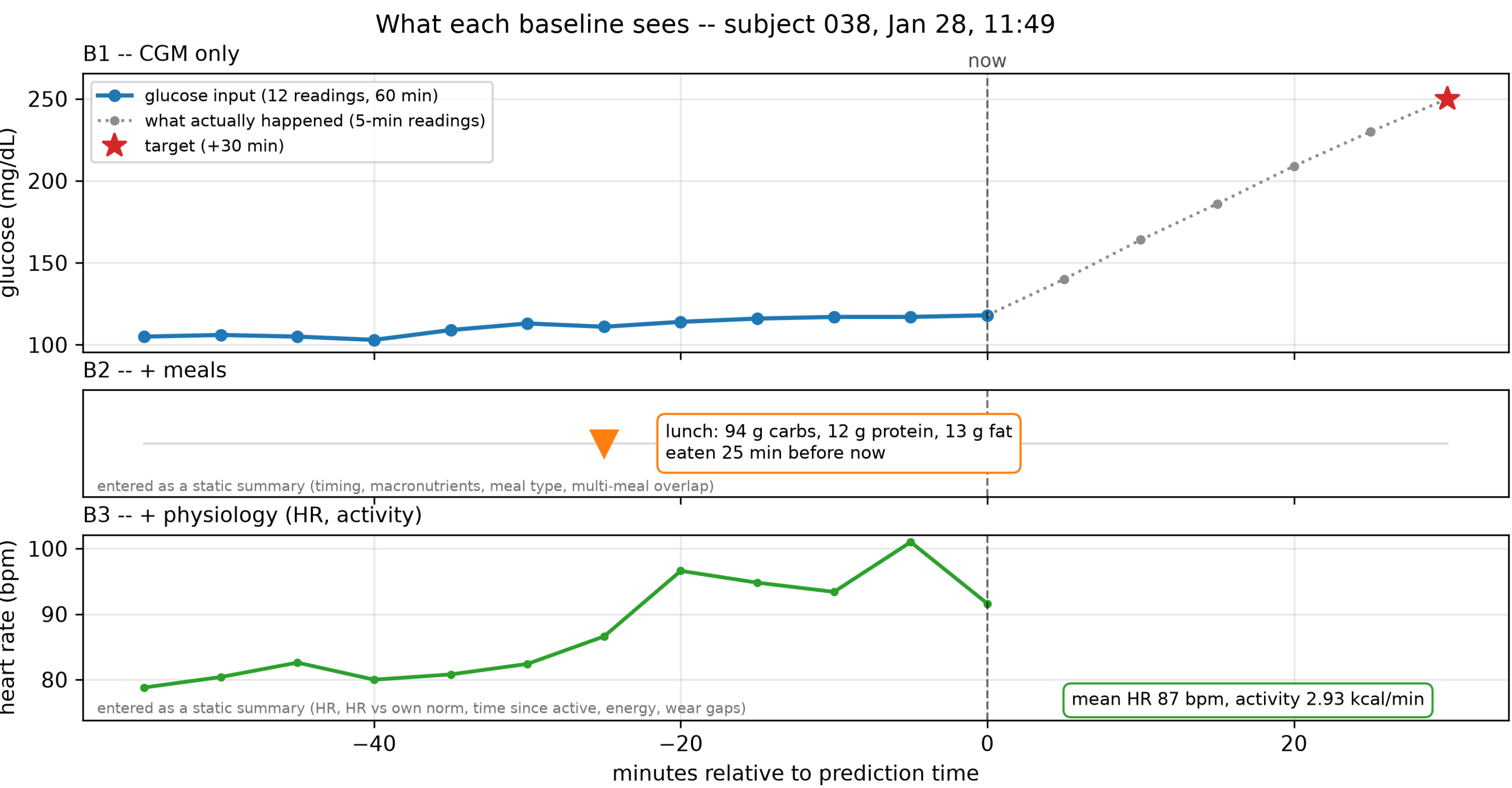
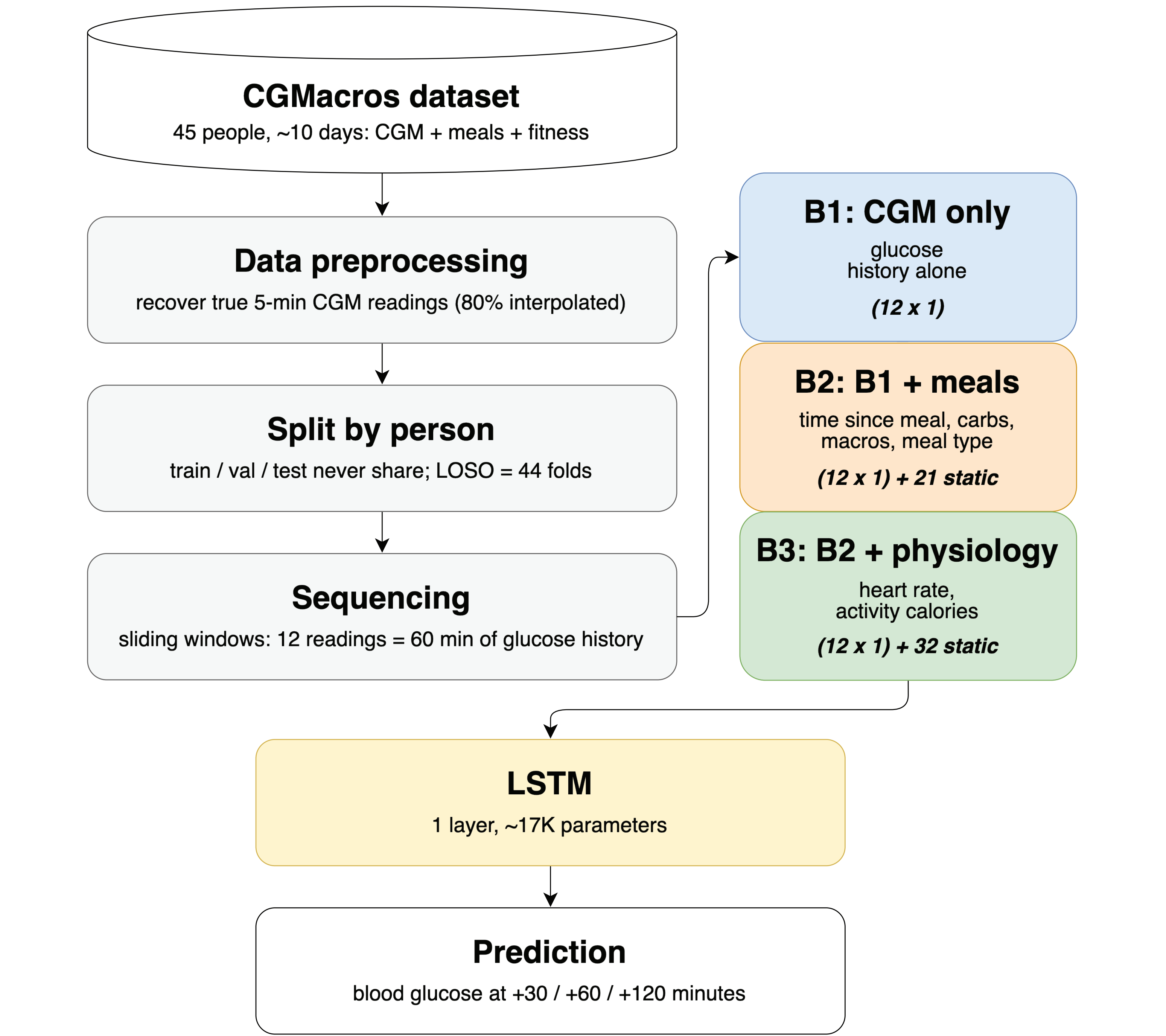
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Motivation & Objective

- Accurate forecasting of future blood glucose can provide early warning of clinically relevant glucose changes.
- Glucose-only forecasting may miss changes associated with food intake and physiological activity.
- This study investigates whether incorporating dietary and multimodal physiological information improves deep-learning-based glucose forecasts.
- LSTM models are evaluated for forecasting horizons of 30–120 min using CGM macros.

Design Flow



Models Tested

- Architectures:** 1-layer, 2-layer, bidirectional, and attention LSTMs.
- Result:** all perform within 0.1 mg/dL – less than random-seed variation.
- Selected:** 1-layer LSTM, 17K parameters; simplest with equivalent performance.
- Training:** 11 random seeds, Adam optimizer, early stopping, fully deterministic.

Results

Post-meal forecast skill: 9.1% → 22.5% (more than doubles).
Dietary information substantially improves forecasting; best gains after meals.

Model	Split	RMSE (mg/dL)	NRMSE
Persistence	pooled	19.19 ± 1.33	0.053
B1, LSTM	pooled	17.69 ± 1.52	0.049
B1, LSTM (2-layer)	val sweep	within 0.04 of 1-layer	n/a
B1, BiLSTM	pooled	17.72 ± 1.28	0.049
B1, Attention LSTM	pooled	17.64 ± 1.39	0.049
B2, LSTM	pooled	16.85 ± 1.17	0.047
B3, LSTM	pooled	16.86 ± 1.19	0.047
Persistence	LOSO	20.12 ± 4.58	0.056
B1, LSTM	LOSO	18.42 ± 4.26	0.051
B2, LSTM	LOSO	17.53 ± 3.83	0.049

Showing forecasts evaluated at 30 minutes ahead for comparison

- NRMSE = RMSE / 360 mg/dL (40–400 mg/dL sensor range)
- Pooled rows average 11 training runs with different random seeds
- LOSO rows average 44 folds, one per held-out subject
- The 2-layer LSTM was compared during tuning on validation data

Key Results

- Dietary info helps:** B2 beats B1 in 11/11 seeds and physiology adds nothing.
- Results hold on unseen subjects:** 12.5% below persistence, 43/44 improve.
- Clinically safer:** 97% predictions in Parkes Zone A vs 93.5% for persistence.
- Better inputs > fancier models:** 4 architectures & hyperparameter sweep all tie.

Prediction Uncertainty

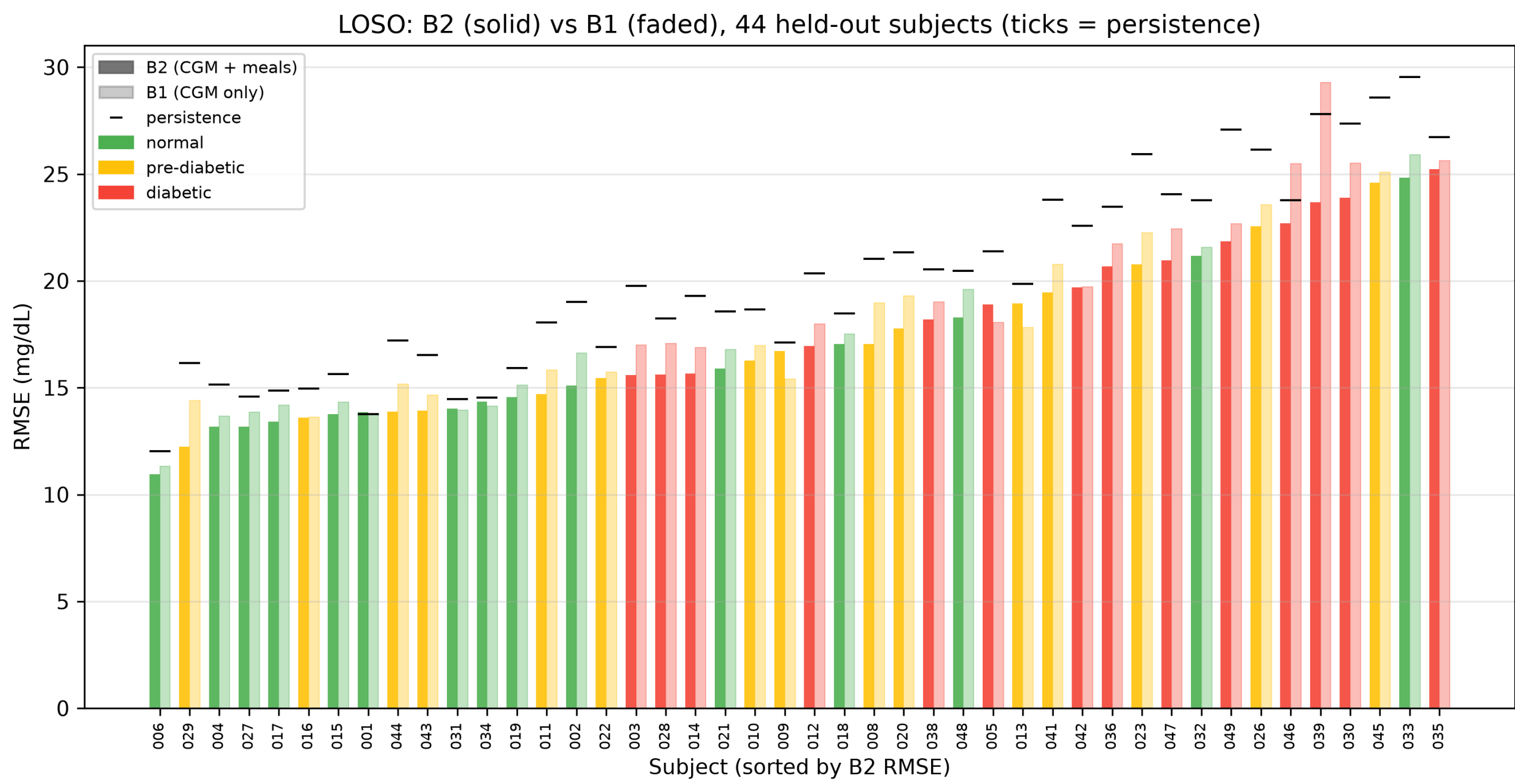
- True glucose values fall within predicted intervals approximately 90% of the time, including for unseen subjects.
- Prediction intervals widen appropriately during rapidly changing glucose conditions: approximately 24 mg/dL when stable versus 49 mg/dL when rising.

Takeaways

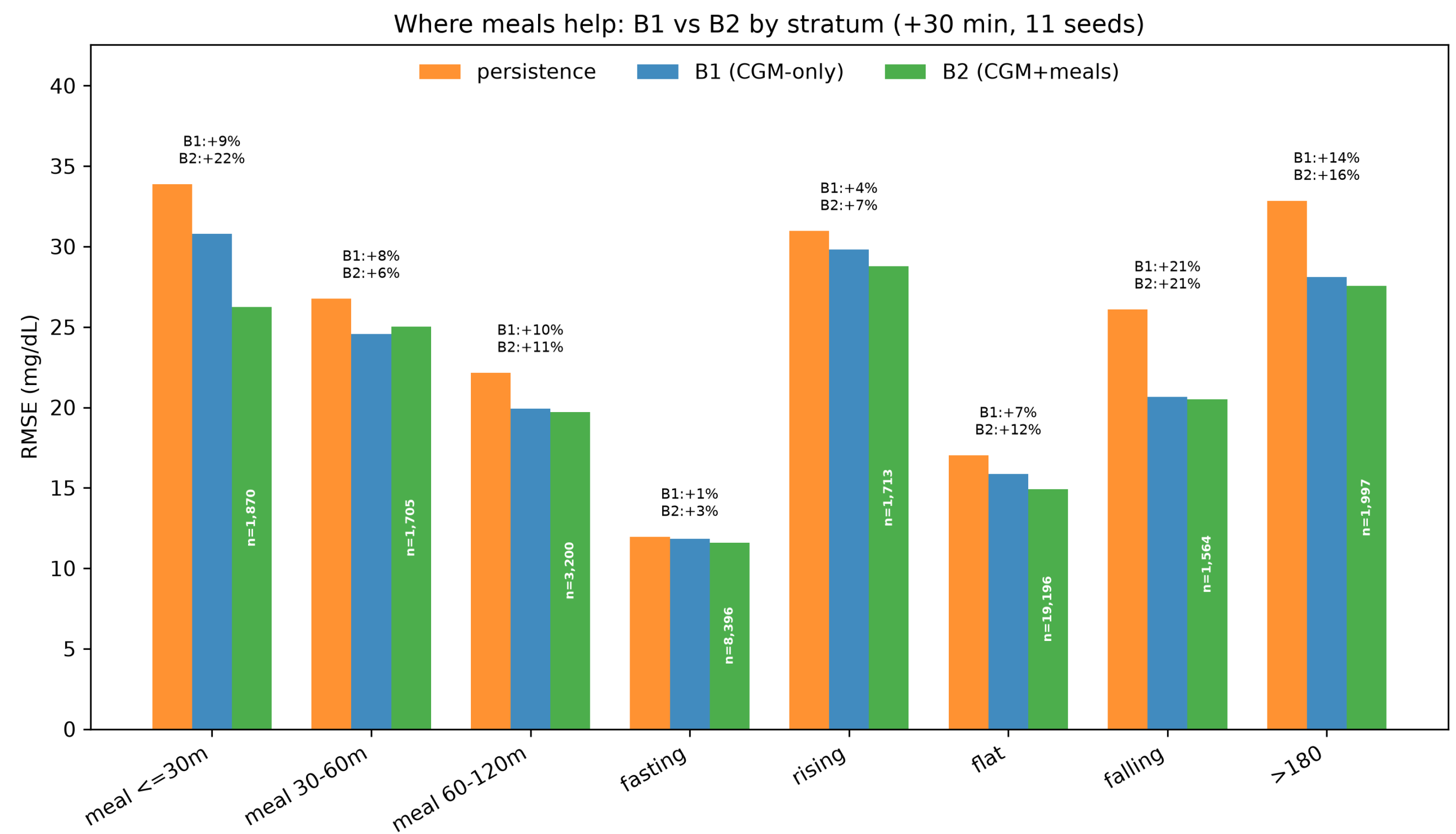
- Forecasting gives 30–60 min of early warning, 97% clinically harmless.
- Meals double forecast skill after eating; diabetic patients benefit most.
- When you ate matters more than what you ate (macros not significant).
- Among macronutrients, only carbohydrates add signal.
- Activity added nothing, a puzzle that may need insulin data to solve.

What Drives Better Forecasts?

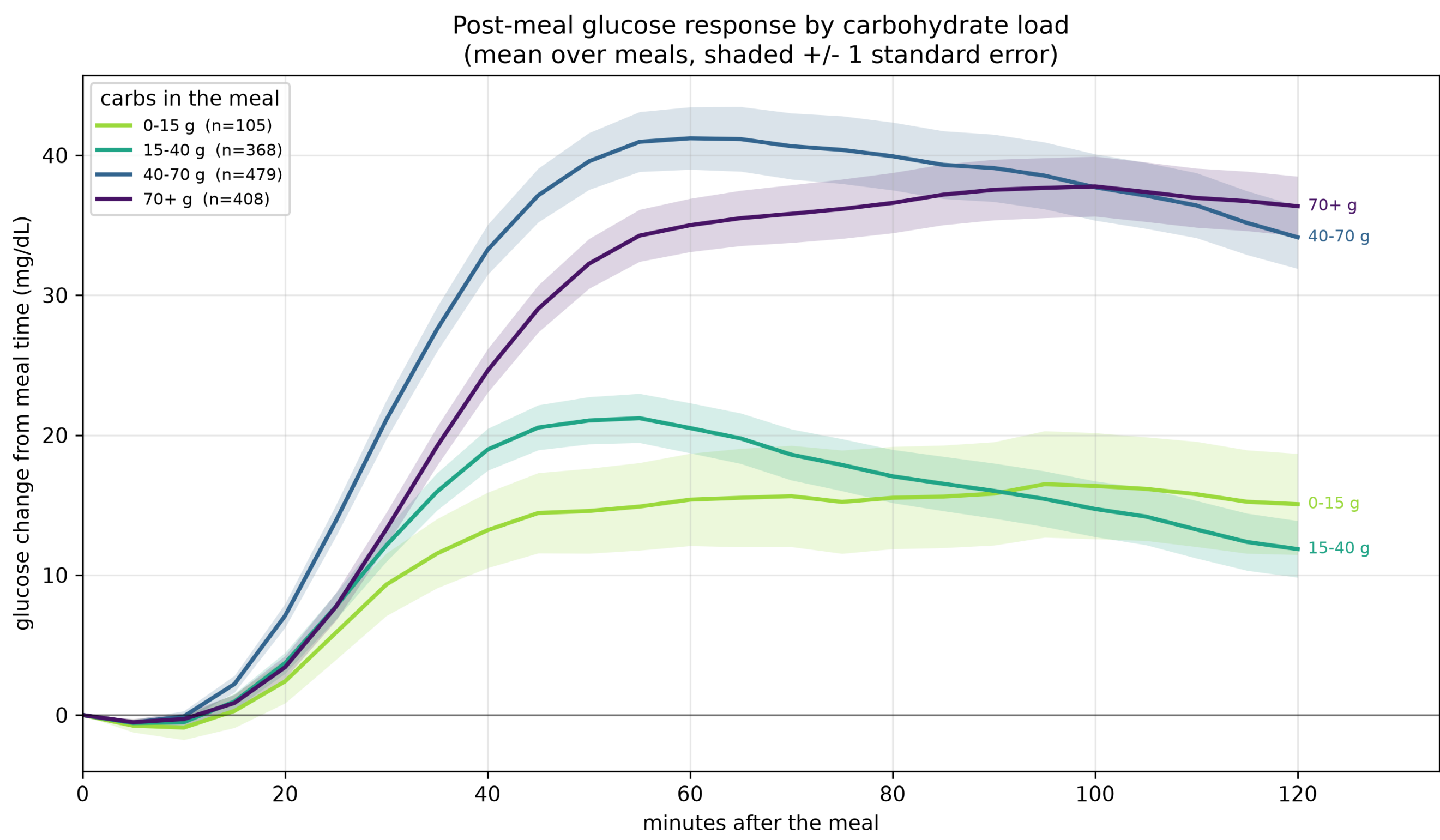
- Dietary information, particularly meal timing and carbohydrate intake, provides the strongest signal, while activity information adds little benefit.



43/44 improve with meals; 039, 046 flip from below to above persistence



Most B2 improvements occur right after meals: 9.1% → 22.5% in skill



Larger carb loads drive bigger rises, saturating around ~40g